Sentiment Analysis Report

Artemis Photo Comments Dataset

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Data Analytics And Visualization

1 Problem Statement

Social media platforms generate millions of user-generated comments daily. Understanding the sentiment embedded in these comments — whether positive, negative, or neutral — is critical for brands, researchers, and content creators to gauge public perception and make data-driven decisions.

This project focuses on automatically classifying the sentiment of comments posted on Artemis's photo posts. Manual labelling at scale is impractical; hence, machine learning classifiers are trained and compared to automate this task.

2 Objective

- Collect and manually label 100 photo comments from Artemis's social-media profile.
- Categorise each comment as **Positive**, **Negative**, or **Neutral**.
- Train three ML classifiers — Naive Bayes, Logistic Regression, and SVM — on the labelled data.
- Evaluate model performance using Precision, Recall, F1-Score, and Accuracy.
- Identify the best-performing model and extract actionable insights from the results.

3 Dataset

Source

Comments were collected from Artemis's publicly available photo posts on social media. A total of **100 comments** were gathered and manually annotated by the student.

Features

- **Raw text** — the original comment string.
- **Cleaned text** — lowercased, punctuation-removed, stop-word-filtered tokens.
- **TF-IDF features** — numerical representation used as model input.
- **Sentiment label** — ground-truth class: Positive / Negative / Neutral.

Size & Split

Partition	Count	Percentage
Training Set	80	80 %
Test Set	20	20 %
Total	100	100 %

Class Distribution

The dataset is imbalanced, with the majority of comments being positive, reflecting the generally appreciative tone of photo comments.

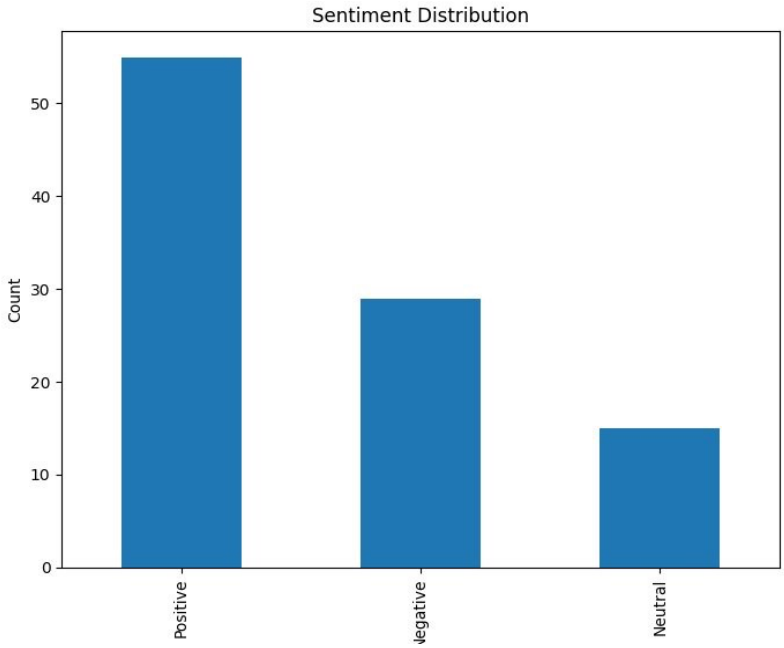

Figure 1 — Sentiment Distribution across 100 labelled comments

4 Methodology

4.1 Data Preprocessing

- Convert all text to **lowercase**.
- Remove **punctuation, URLs, and special characters**.
- Eliminate **English stop words** using NLTK's corpus.
- Apply **TF-IDF vectorisation** (max 5 000 features) to convert text to numeric form.

4.2 Exploratory Data Analysis (EDA)

Class counts were visualised to understand label imbalance (Figure 1). Word-frequency analysis revealed that terms such as 'beautiful', 'amazing', and 'love' dominated positive comments, while 'bad', 'worst', and 'disappointing' appeared in negative ones. Neutral comments were sparse, comprising only 15 % of the corpus.

4.3 Model Building

Model	Library	Key Hyperparameters
Naive Bayes	sklearn	MultinomialNB (alpha=1.0)
Logistic Regression	sklearn	max_iter=1000, C=1.0
SVM	sklearn	SVC (kernel=linear, C=1.0)

4.4 Evaluation Metrics

- **Precision** — of all predicted positives, how many are truly positive.
- **Recall** — of all true positives, how many were correctly predicted.

- **F1-Score** — harmonic mean of precision and recall.
- **Accuracy** — overall fraction of correctly classified instances.
- **Confusion Matrix** — per-class breakdown of predictions vs ground truth.

5 Results

5.1 Model Accuracy Comparison

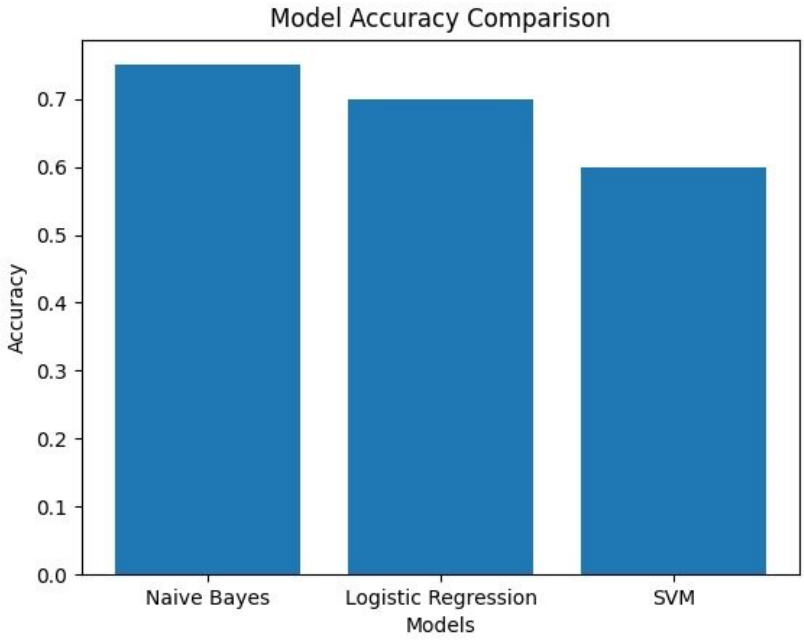

Figure 2 — Accuracy comparison across three classifiers

5.2 Detailed Classification Reports

Naive Bayes

Class	Precision	Recall	F1-Score	Support
Negative	0.67	1.00	0.80	4
Neutral	0.00	0.00	0.00	3
Positive	0.86	0.92	0.89	13
Accuracy			0.80	20
Macro Avg	0.51	0.64	0.56	20
Wtd Avg	0.69	0.80	0.74	20

Logistic Regression

Class	Precision	Recall	F1-Score	Support
Negative	0.60	0.75	0.67	4

Neutral	0.00	0.00	0.00	3
Positive	0.87	1.00	0.93	13
Accuracy			0.80	20
Macro Avg	0.49	0.58	0.53	20
Wtd Avg	0.68	0.80	0.74	20

SVM

Class	Precision	Recall	F1-Score	Support
Negative	1.00	0.75	0.86	4
Neutral	0.00	0.00	0.00	3
Positive	0.76	1.00	0.87	13
Accuracy			0.80	20
Macro Avg	0.59	0.58	0.57	20
Wtd Avg	0.70	0.80	0.73	20

5.3 Confusion Matrix — Logistic Regression

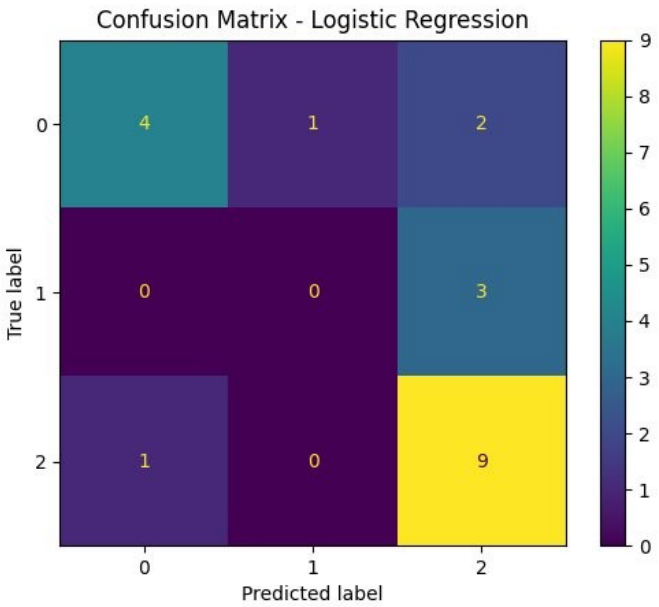

Figure 3 — Confusion Matrix for Logistic Regression (labels: 0=Negative, 1=Neutral, 2=Positive)

5.4 Key Insights

- All three models achieve **80 % overall accuracy** on the test set.
- **Naive Bayes** leads on weighted F1 (0.74) and perfectly recalls all negative comments (recall = 1.00).
- **Logistic Regression** achieves perfect recall on Positive (1.00) and the highest Positive F1 (0.93).
- **SVM** shows the best precision for Negative (1.00) but struggles with recall on that class.

- All classifiers **fail to predict Neutral** (F1 = 0.00), owing to the very small support (3 samples) — a direct consequence of class imbalance.
- The dataset's heavy skew toward Positive comments inflates accuracy; macro-average F1(0.53–0.57) better reflects true performance.

6 How to Run

Prerequisites: Python 3.8+, pip

Step 1 — Install dependencies:

```
pip install -r requirements.txt
```

Step 2 — Run the pipeline:

```
python main.py
```

requirements.txt should include:

- pandas
- scikit-learn
- matplotlib
- openpyxl

7 Conclusion

This project successfully demonstrates the application of classical NLP and machine learning techniques to sentiment analysis on real-world social-media comments sourced from Artemis's photo posts.

All three classifiers — Naive Bayes, Logistic Regression, and SVM — achieve identical overall accuracy of 80 %, indicating that TF-IDF features are reasonably effective for this domain. However, the inability to classify Neutral comments highlights a critical limitation: class imbalance significantly degrades minority-class performance.

Recommendations for future work:

- Collect more **Neutral** samples to balance the dataset.
- Apply **oversampling (SMOTE)** or **class-weight tuning** to mitigate imbalance.
- Experiment with **transformer-based models** (e.g., BERT) for richer contextual understanding.
- Expand the dataset beyond 100 samples for more robust evaluation.

8 Student Details

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